

Mediterranean diet, cognitive function, and dementia: a systematic review.

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BACKGROUND: Adherence to a Mediterranean diet has been associated with lower risk of various age-related diseases including dementia. Although narrative reviews have been published, no systematic review has synthesized studies on the association between Mediterranean diet adherence and cognitive function or dementia. METHODS: We conducted a systematic review of 11 electronic databases (including Medline) of published articles up to January 2012. Reference lists, selected journal contents, and relevant websites were also searched. Study selection, data extraction, and quality assessment were performed independently by two reviewers using predefined criteria. Studies were included if they examined the association between a Mediterranean diet adherence score and cognitive function or dementia. RESULTS: Twelve eligible papers (11 observational studies and one randomized controlled trial) were identified, describing seven unique cohorts. Despite methodological heterogeneity and limited statistical power in some studies, there was a reasonably consistent pattern of associations. Higher adherence to Mediterranean diet was associated with better cognitive function, lower rates of cognitive decline, and reduced risk of Alzheimer disease in nine out of 12 studies, whereas results for mild cognitive impairment were inconsistent. CONCLUSIONS: Published studies suggest that greater adherence to Mediterranean diet is associated with slower cognitive decline and lower risk of developing Alzheimer disease. Further studies would be useful to clarify the association with mild cognitive impairment and vascular dementia. Long-term randomized controlled trials promoting a Mediterranean diet may help establish whether improved adherence helps to prevent or delay the onset of Alzheimer disease and dementia.